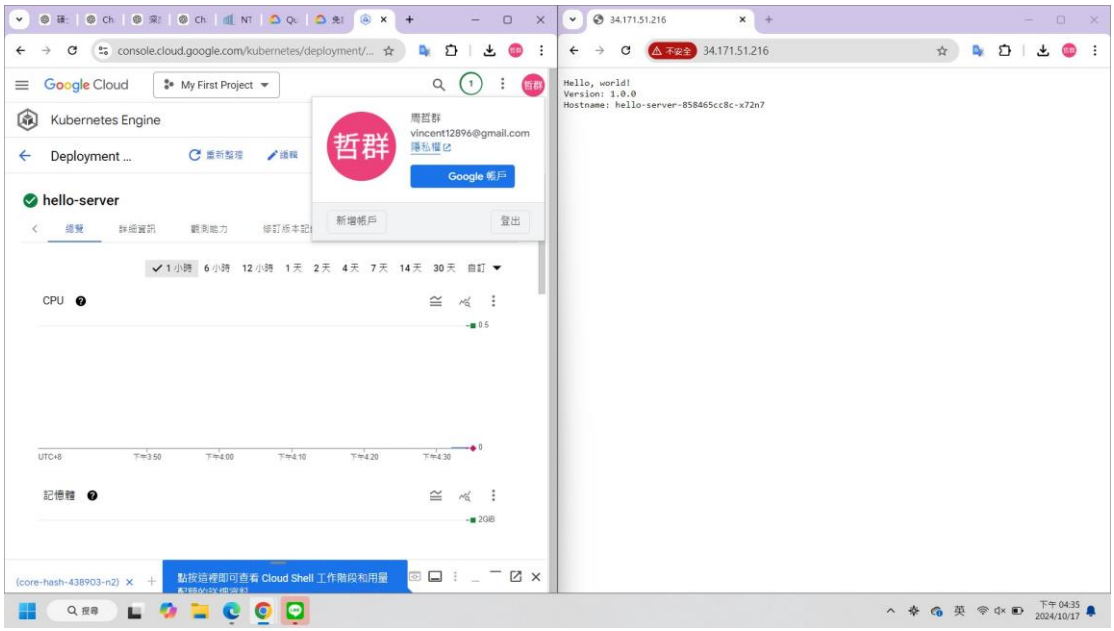
LAB1



Lab1 做的事情為如何將一個簡單的容器化應用程式部署到 Google Kubernetes Engine (GKE) 叢集中，並讓外部使用者可以存取這個應用程式。

大致的步驟如下：

1.建立 GKE cluster

```
gcloud container clusters create-auto hello-cluster \
    --location=us-central1
```

2.取得 cluster 的認證憑證

```
gcloud container clusters get-credentials hello-cluster \
    --location us-central1
```

3.部屬應用程式(包含建立和公開 deployment)

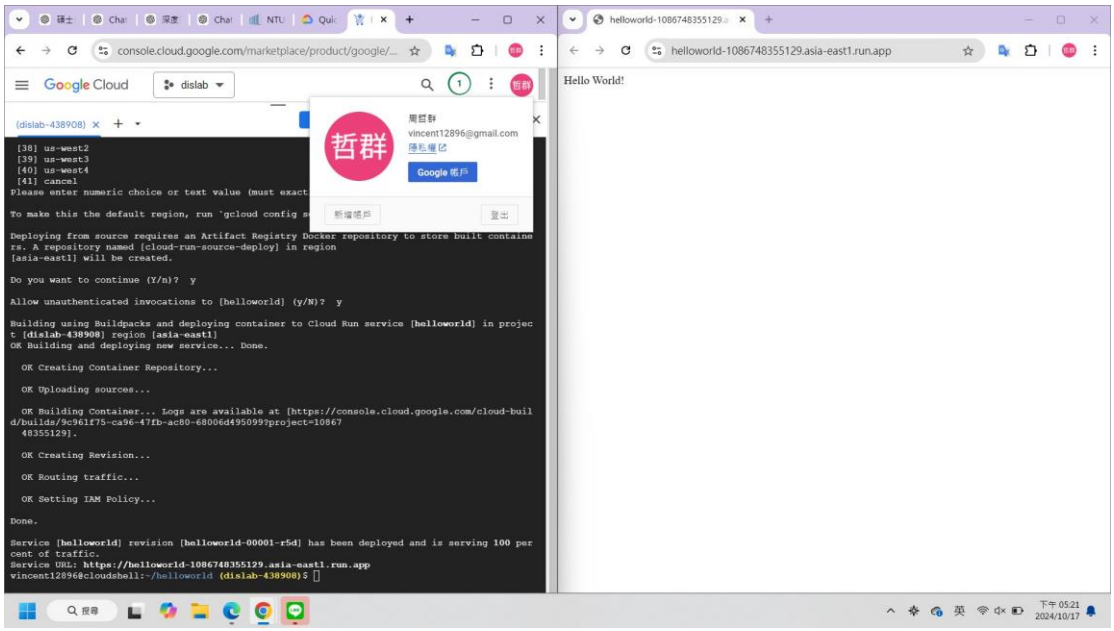
```
kubectl create deployment hello-server \
    --image=us-docker.pkg.dev/google-samples/containers/gke/he
```

```
kubectl expose deployment hello-server \
    --type LoadBalancer \
    --port 80 \
    --target-port 8080
```

4.最後檢查是否能透過 IP 取得外部應用程式

```
kubectl get service hello-server
```

LAB2



Lab2 如何將一個 Node.js 的 "Hello World" 服務容器化，並將其部署到 Google Cloud Run。

大致的步驟如下：

1.啟用 API 並給 Compute Engine 預設服務帳號 Cloud Build 角色

```
gcloud services enable run.googleapis.com \
    cloudbuild.googleapis.com
```

```
gcloud projects add-iam-policy-binding PROJECT_ID \
    --member=serviceAccount:PROJECT_NUMBER-compute@deve
    --role=roles/cloudbuild.builds.builder
```

2.建立資料夾並創建 package.json 跟 index.js 兩個檔案

```
mkdir helloworld
cd helloworld
```

```
{
  "name": "helloworld",
  "description": "Simple hello world sample in Node",
  "version": "1.0.0",
  "private": true,
  "main": "index.js",
  "type": "module",
  "scripts": {
    "start": "node index.js"
  },
  "engines": {
    "node": ">=16.0.0"
  },
  "author": "Google LLC",
  "license": "Apache-2.0",
  "dependencies": {
    "express": "^4.17.1"
  }
}
```

```
import express from 'express';
const app = express();

app.get('/', (req, res) => {
  const name = process.env.NAME || 'World';
  res.send(`Hello ${name}!`);
});

const port = parseInt(process.env.PORT) || 8080;
app.listen(port, () => {
  console.log(`helloworld: listening on port ${port}`);
});
```

3.最後一樣部屬並檢查服務

```
gcloud run deploy --source .
```